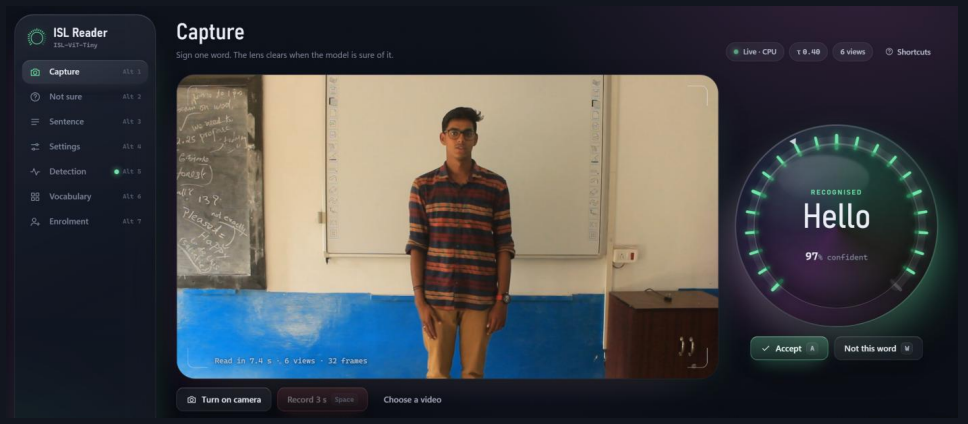
ISL Reader

A 2 MB Vision Transformer, a super-resolution GAN front end and an integrated interface for Indian Sign Language



Team 3

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Guide: Dr. T. Bagyammal · 23CSE473 NNDL · Review 3

What we set out to build

A model of about 2 MB

It has to run on smart glasses, so the classifier must fit in roughly two megabytes.

≥ 75 % on unseen people

Scored only on recording sessions the model never saw, over the 50 deployed words.

A working app

Camera or video in, word out, with a confidence check and a way to correct the answer.

Why it is hard

- Most signs use both hands, and handshape, motion and location together decide the word.
- Labelled data is scarce: INCLUDE has about 15 clips per word.
- A wearable camera sends smaller, more compressed video than the 1080p studio footage.
- Random splits leak recording sessions, which inflates published accuracies.

Dataset details

4,257
INCLUDE clips used

262
words, 15 categories

1080p
25 fps, one studio

13
recording sessions

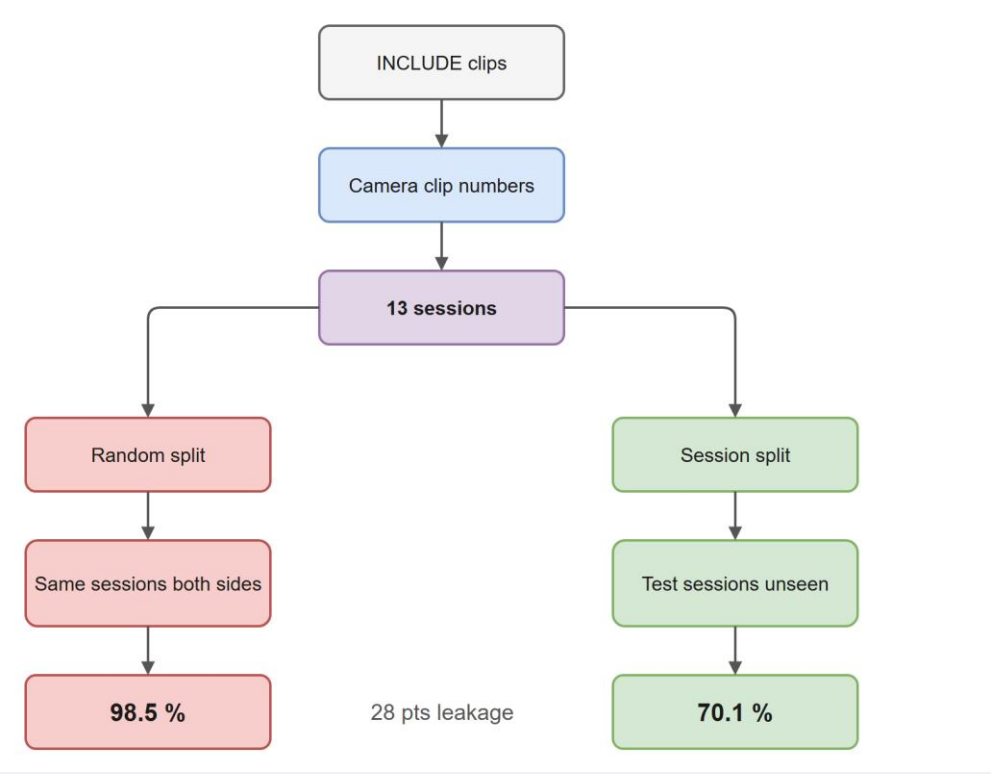
Split	Clips	Use
Training sessions	2,845	Classifier and GAN training
Held-out test sessions	1,010	472 of them = the 50 deployed words
Excluded	402	Words with < 6 training clips
Validation sessions	888	Choosing between GAN v3 arms
iSign (unlabelled)	18,000	Self-supervised pretraining; 17,896 kept
CISLR (300 x 300 px)	609	Cross-corpus test

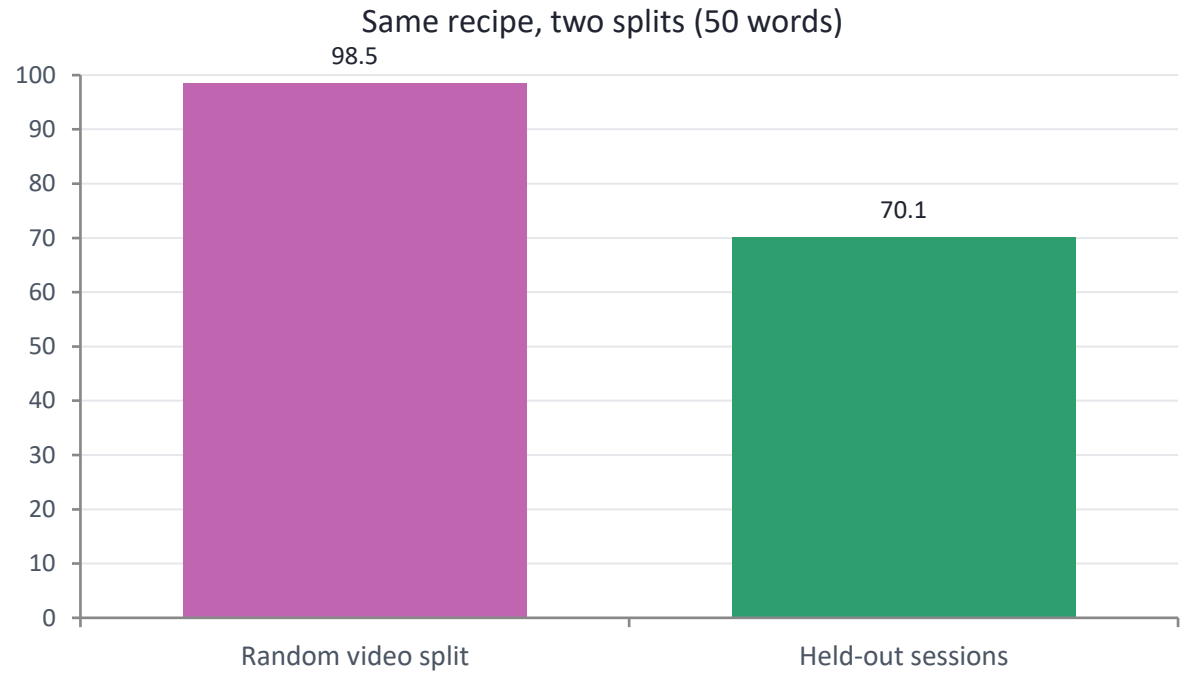


Frames from held-out INCLUDE clips (Sridhar et al., ACM MM 2020, CC-BY-4.0)

We train on all 262 words and answer from the 50 deployed ones. Each word has a fixed number of clips, so training on fewer words would only discard data.

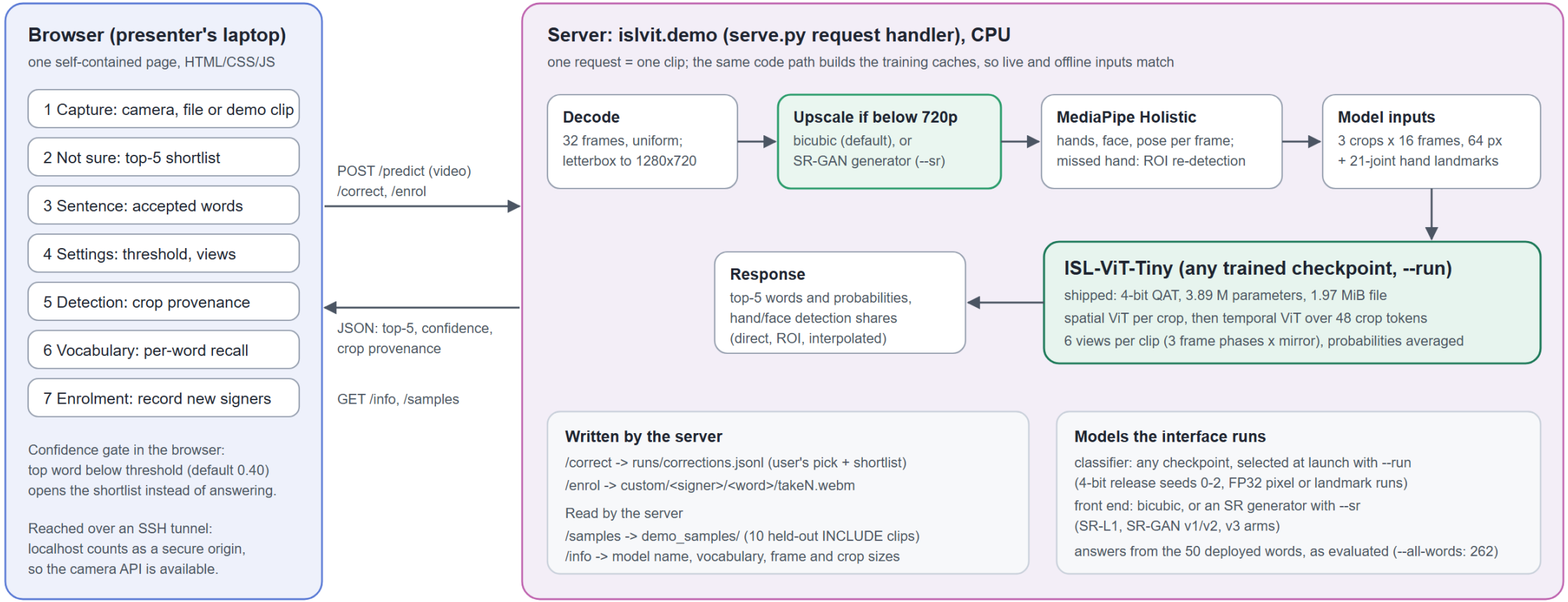
Evaluation: hold out whole recording sessions



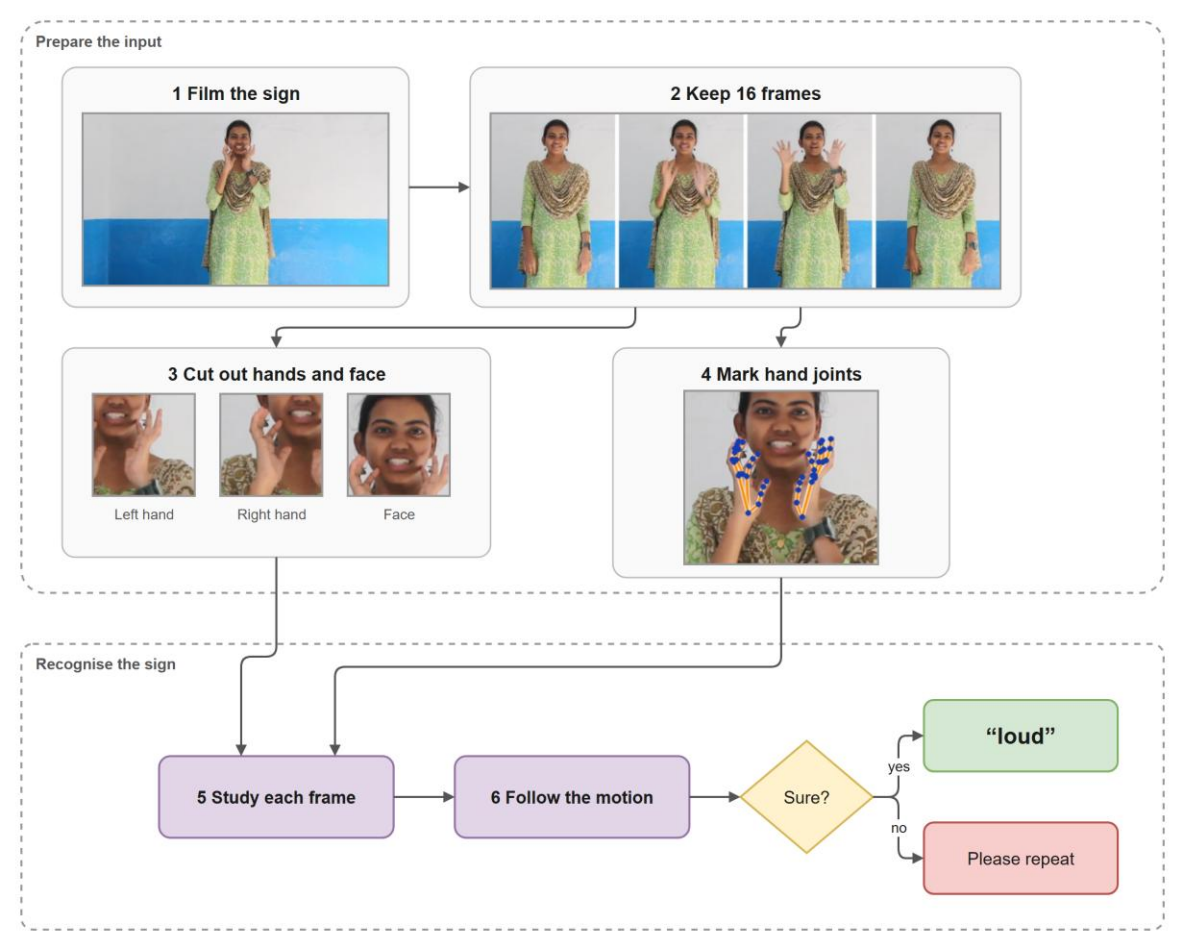


- Camera clip numbers run in sequence, and a large jump marks a new session.
- Whole sessions are held out, and the split builder rejects any overlap.
- The 28-point gap is what the model gains from having seen the same person and room.

System architecture of the integrated app

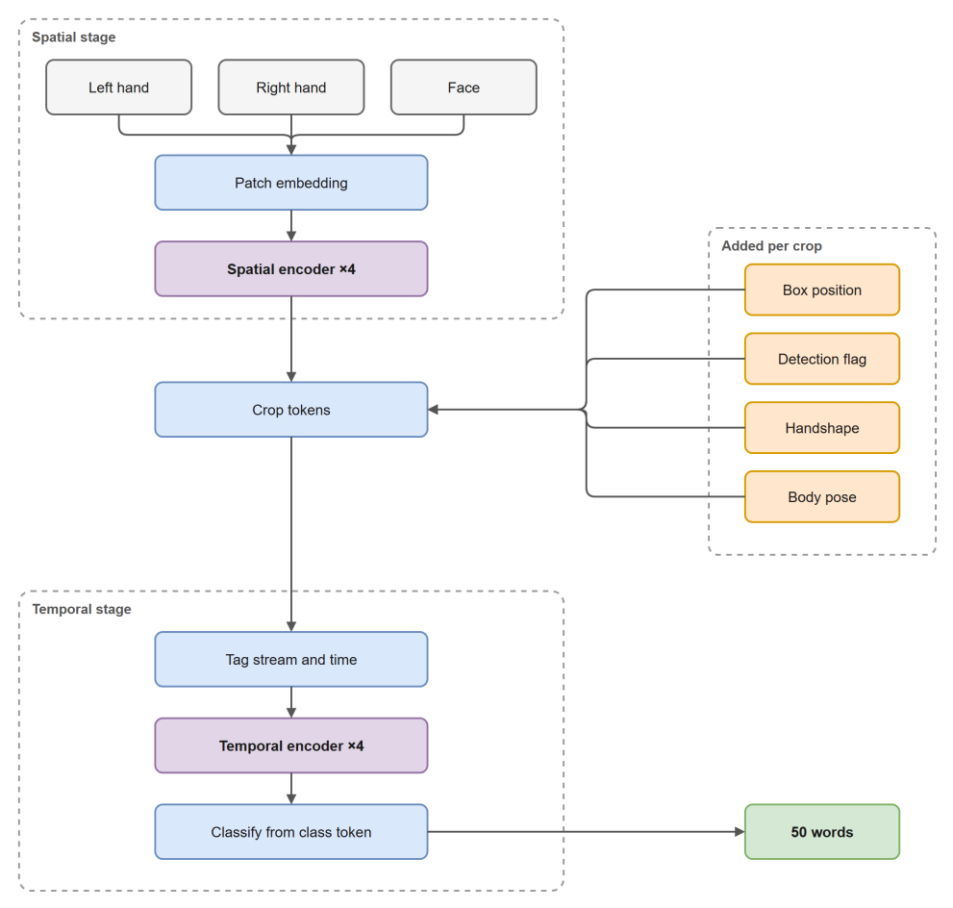


From video to model inputs



- 32 frames sampled uniformly and letterboxed to 1280 × 720.
- MediaPipe Holistic finds the hands, face and pose. A missed hand gets a second search in an upscaled forearm region; failing that, the nearest box is reused and flagged.
- Three 64 px crops per frame: left hand, right hand, face.
- 21 joints per hand: shape measured from the wrist, location measured from the shoulders.
- Detectors are rebuilt for every video. Without that, repeated runs disagreed on 3–6 % of frames.

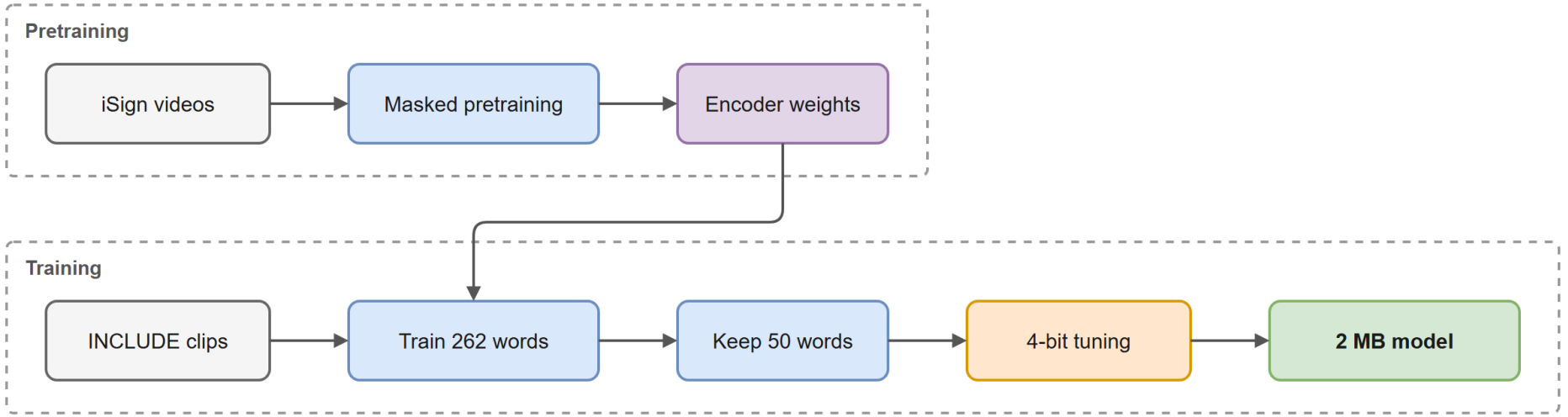
Classifier: ISL-ViT-Tiny



Component	Setting
Input	16 frames x 3 crops x 64 x 64, box geometry, detection flags, landmarks
Spatial stage	4 blocks, width 192, 3 heads, shared over 48 crops
Temporal stage	4 blocks over 48 crop tokens + class token
Output	262 words, restricted to the 50 deployed
Parameters	3,894,214 (91 k for landmarks)

Why factorised: attending over every patch at once would take 816 tokens. The model attends over 17 tokens inside each crop, then 49 across the clip. Box geometry restores the location that cropping throws away.

Training algorithm and compression



1. Self-supervised pretraining

Masked image modelling (SimMIM) on 17,896 of 18,000 unlabelled iSign clips: 75 % of patches are hidden and reconstructed.

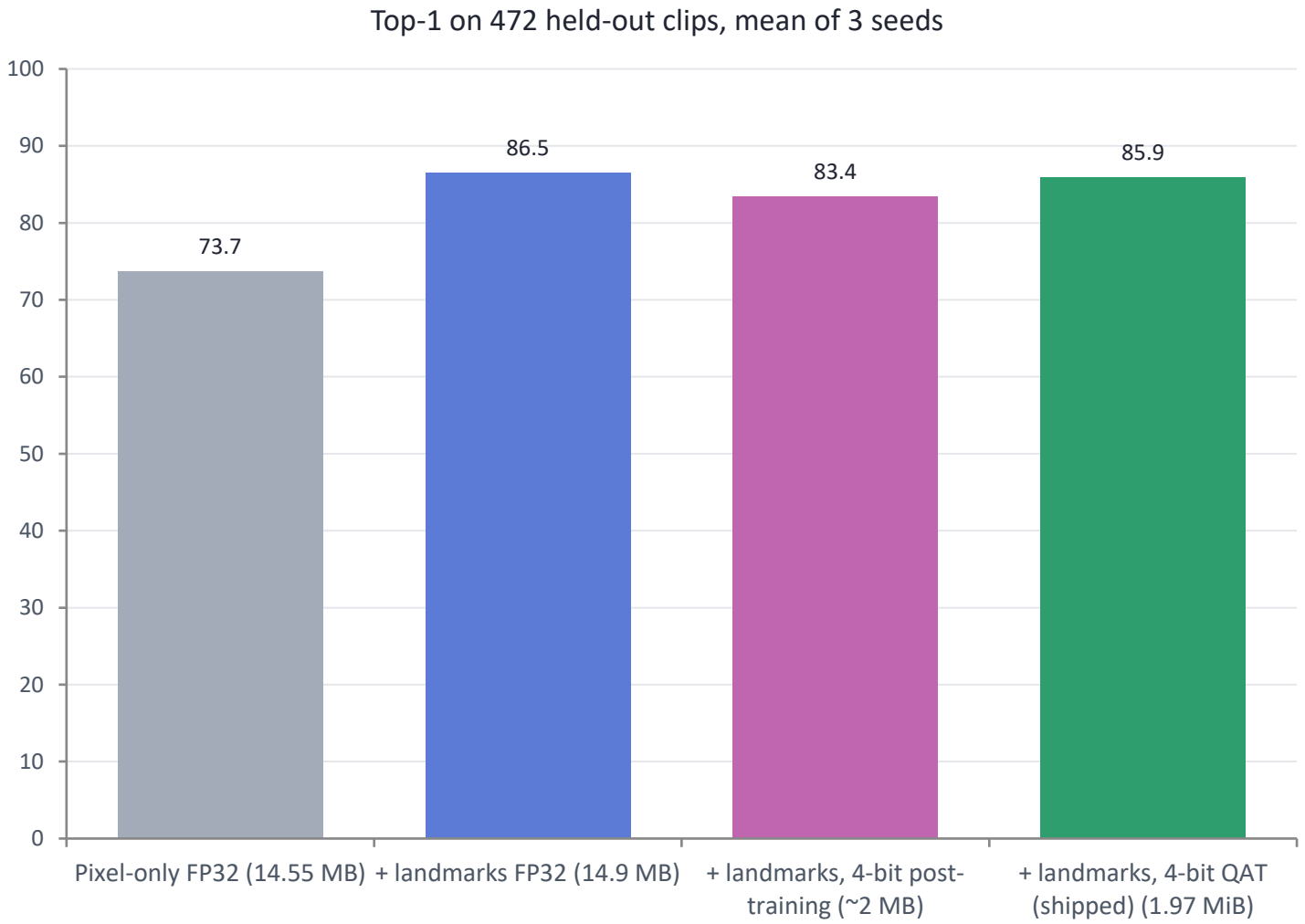
2. Supervised training

1,000 epochs on 2,845 clips with AdamW 5e-4, mixup, stream dropout, resolution jitter and EMA. We keep the final epoch, so nothing is selected on test.

3. 4-bit quantisation-aware

4-bit weights in groups of 128 with 16-bit scales, a straight-through estimator, 300 epochs. Rounding after training lost 3.1 points; QAT lost 0.6.

Comparing the classifier variants



85.9 %

top-1, shipped 4-bit model (seeds 84.5 / 87.5 / 85.8)

96.4 %

top-5: the right word is usually in the shortlist

1.97 MiB

3.89 M parameters, 4-bit

Answering from all 262 words instead of the 50 gives 72.0 %, so the app restricts its answers the same way the evaluation does.

Where a GAN can help: low-resolution video



Why here

Every clip reaches the hand detector through one resize to 720p, which used bicubic. Small, compressed video from a wearable camera would pass through exactly this step. The GAN replaces only that resize, and 1080p video never reaches it, so clean results cannot change.

How we tested it

472 test clips degraded to 360p, 240p and 180p with JPEG. Each GAN is compared with bicubic and with a no-GAN control: SR-L1 for v1 and v2, v3 without the GAN for v3. Frozen 4-bit and 32-bit models, plus models fine-tuned on 360p. Paired McNemar tests under a rule fixed in advance.

3

levels: 360p, 240p, 180p, each with JPEG

6

front ends: bicubic, SR-L1, SR-GAN v1, v2, two v3 arms

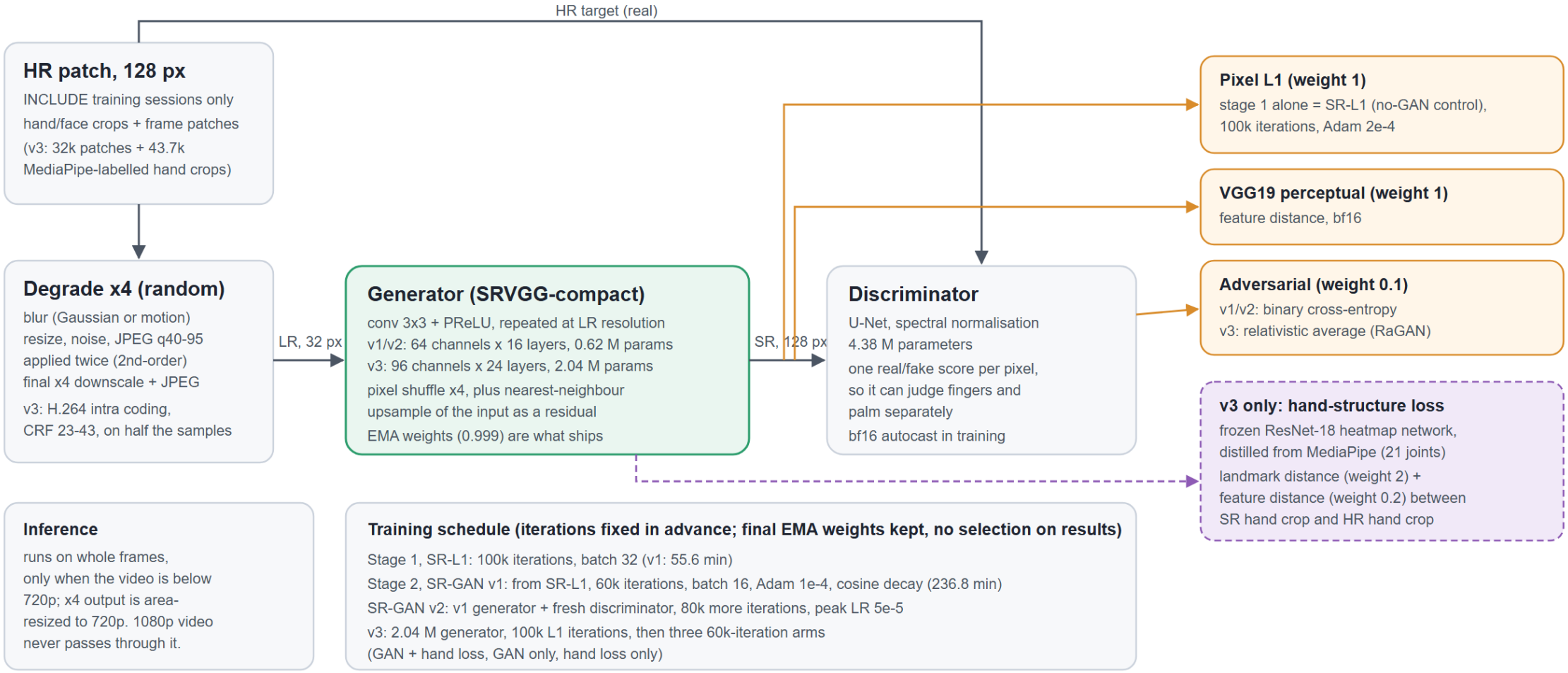
472

test clips, identical in every run

$p < 0.05$

and all 3 seeds agree, fixed in advance

The super-resolution GAN



Three GAN versions, each built for a reason

Version	What changed	Why we built it	Against its no-GAN control
SR-GAN v1	Real-ESRGAN recipe: L1 + VGG19 + adversarial (BCE), 60,000 iterations from SR-L1; 0.62 M-parameter generator	The planned design	No reliable difference at 360p, 240p or 180p
SR-GAN v2	Same generator, new discriminator, 80,000 more iterations at half the learning rate	v1's discriminator dominated: was v1 undertrained?	Better only at 240p (+1.5); 1.2 points worse than v1 at 360p
SR-GAN v3	2.04 M-parameter generator, hand-landmark loss, relativistic adversarial loss, H.264 in training	The remaining error is in the hand landmarks	Better only at 240p (+2.4); found the most hands (77 % at 180p)

2 of 9

GAN vs no-GAN tests passed (3 versions × 3 levels), both at 240p

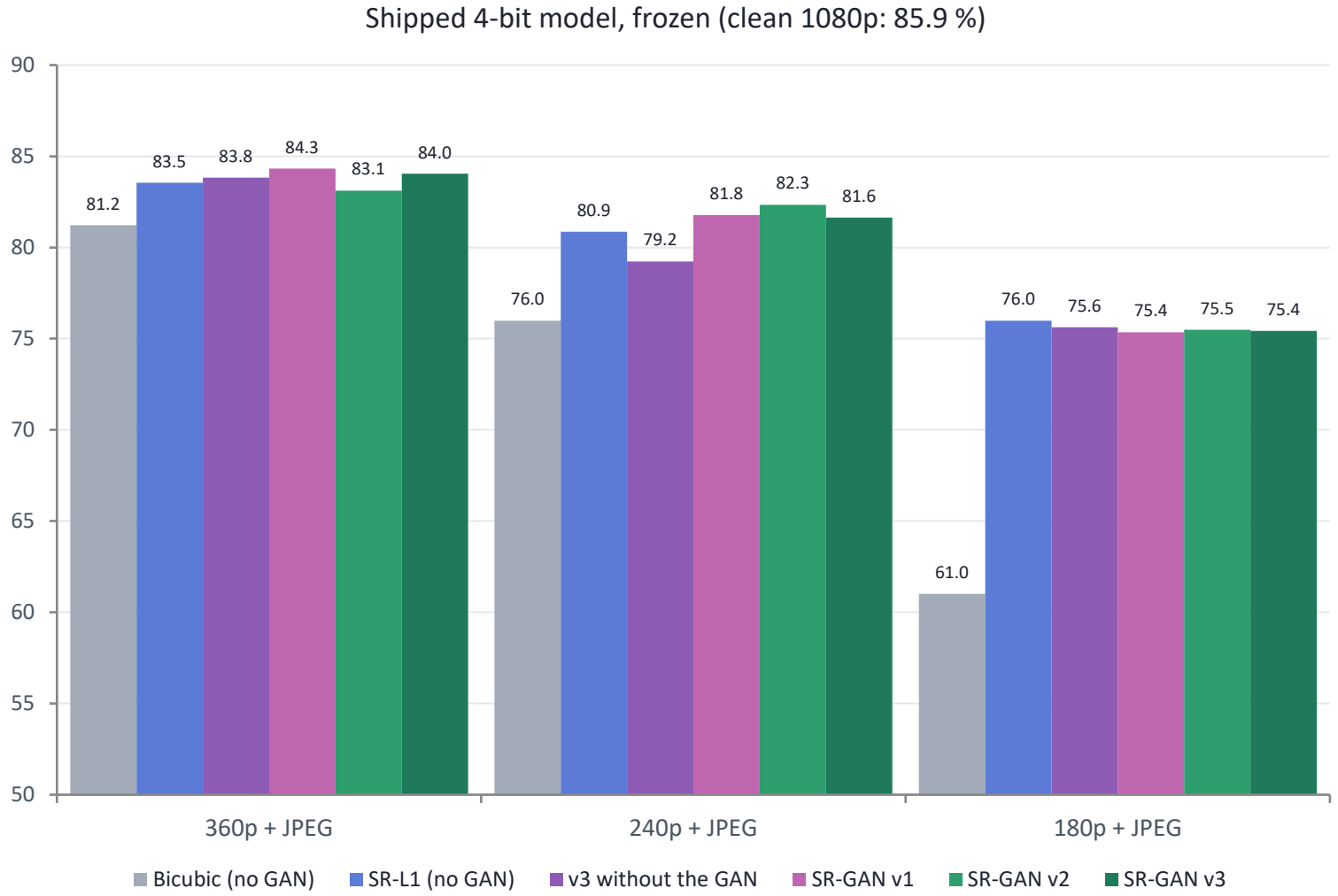
−1.2

points for v2 against v1 at 360p ($p = 0.021$): training longer did not help

15.5 %

of hands missed at 180p with v3, against 22.1 % with v2 and 39.3 % with bicubic

GAN results on degraded test video



+14.5

points at 180p for SR-GAN v2 over bicubic ($p = 7 \times 10^{-32}$)

+15.0

points at 180p for SR-L1, which has no GAN

≤ 1 point

between any two learned upscalers at 180p

Fine-tuned on 360p, v1 and v2 still beat bicubic by 5.2–5.7 points at 240p and 12.5–13.3 at 180p.

Where the remaining loss sits



Landmarks, not pixels

With SR-GAN v2, 10.4 points are still missing at 180p: about 7.7 on the landmark side and 2.8 on the pixel side.

Detector tuning does not help

Lower thresholds and wider searches found up to 99 % of hand frames, yet accuracy moved at most 0.8 points, because misplaced hands rose to 14 %.

Few invented hands

Hands found only in degraded video at 180p: 1.9 % with bicubic, 2.6 % with SR-GAN v2, 3.1 % with v3.

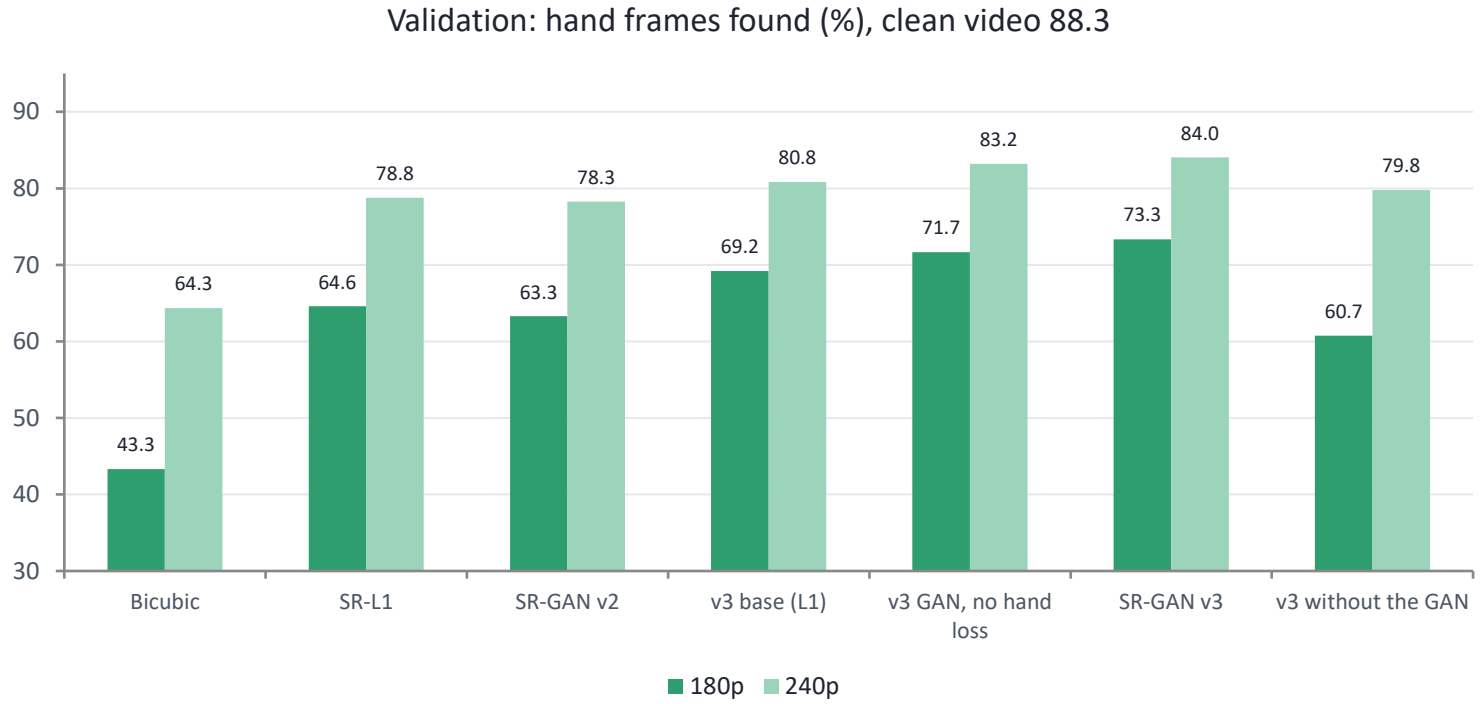
SR-GAN v3: trained to keep the hand joints

Hand-landmark proxy

A ResNet-18 with a heatmap head, trained on 43,682 verified hand crops to read MediaPipe's 21 joints.

On validation crops its error is 0.0215 of the crop, with 91.4 % of points within 0.05 (MediaPipe against itself: 0.041).

The generator is penalised when the proxy reads different joints in its output than in the original.

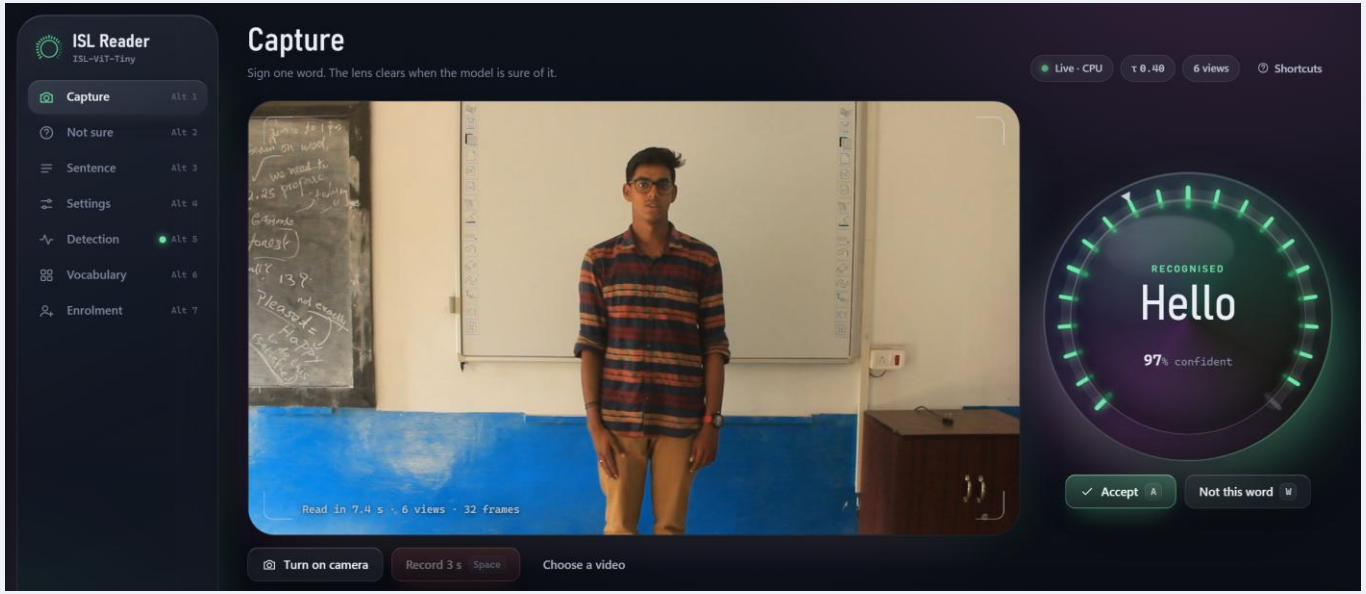


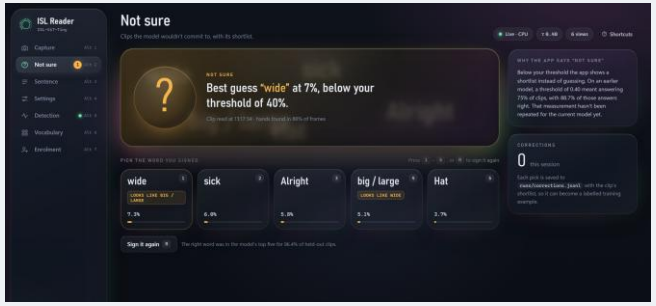
On validation it found more hands (73.3 % vs 63.3 % for v2 at 180p). On the test set it was not reliably better than v2 or SR-L1, and v3 without the GAN was 3.1 points worse than v2 at 240p. Better detection did not become better classification.

−3.1

points for v3 without the GAN against v2 at 240p ($p = 5 \times 10^{-6}$)

The interface, connected to the models







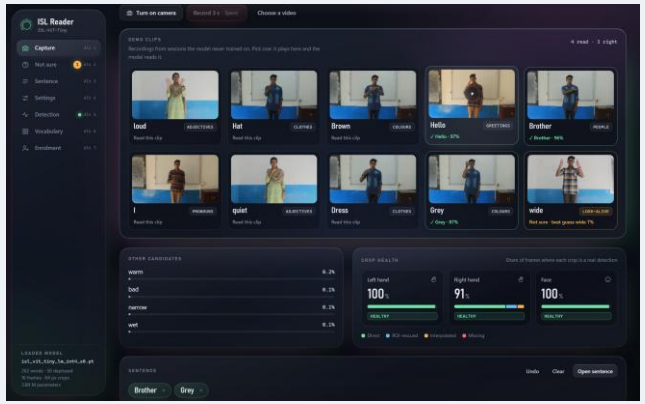
Seven screens

- Capture: camera, video file or one of 10 held-out demo clips
- Not sure: shortlist when confidence < threshold
- Sentence: accepted words, speak, copy, export
- Settings: threshold and views
- Detection: where each crop came from
- Vocabulary: per-word recall on unseen sessions
- Enrolment: record new signers for training

Runs any trained checkpoint behind either upscaling front end. It is reached over an SSH tunnel, which also lets the browser use the camera.

Testing the interface with the models

Test	Method	Result
Screens render	Chromium, 7 screens at 1366 and 1920 px	No console errors
Live answers	10 demo clips on the live server; 4 also via the Capture screen	9/9 reliable clips correct (66–97 %); look-alike → Not sure
Same answers as the evaluation	App vs offline result, same clips and model	8 of 8 clips agree
Vocabulary restriction	50-word vs 262-word answers	Matches the 85.9 % setting (262 words: 72.0 %)
GAN front end in the app	180p copy of a demo clip, SR-GAN on	Read correctly
Latency	Desktop CPU, 6 views, end to end	5–9 s per clip





What we found

- 85.9 % top-1 on unseen sessions from a 1.97 MiB model. Under the same recipe, a random split scored about 28 points higher.
- A learned upscaler recovers up to 15 points on 180p video. Across three GAN versions, the adversarial loss beat its no-GAN control in 2 of 9 tests.
- After upscaling, most of the remaining error is in where the hand joints are placed, not in the pixels.
- The app gives the same answers as the offline evaluation on every clip checked.

Next steps

- Fine-tune the classifier on SR-GAN v3's output
- Re-measure the confidence curve for the shipped model
- Build a detection path for tightly framed video (CISLR)
- Record new signers through Enrolment; test on wearable hardware

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